Cite this paper: Dai, J.; Duan, X.; Zhou, J.; Fu, C.; Ma, S. *Chin. J. Chem.* **2018**, *36*, 387—391.

Breaking Report

DOI: 10.1002/cjoc.201800090

Catalytic Enantioselective Simultaneous Control of Axial Chirality and Central Chirality in Allenes†

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ABSTRACT Chiral molecules, which may contain one or more different type(s) of stereocentres, such as central, axial, planar, and helical chiralities, *etc*., are indispensable in chemistry, pharmaceutical industry, and life science. Despite many advances for the preparation of chiral molecules usually with a single type of chirality have been realized, simultaneous construction of different types of chiralities is still a significant challenge. Here, we wish to report a protocol for preparation of chiral allenes with both central and axial chiralities via a catalytic asymmetric allenylation of different biologically or synthetically useful fluorinated or non-fluorinated nucleophiles with readily available racemic allenes by using a single chiral ligand. An echoing between the central chirality and axial chirality for the enantioselectivity was observed. This strategy provides a general and practical approach to functionalized optically active allenes bearing both central and axial chiralities with an excellent enantioselectivity under mild conditions.

KEYWORDS chiral palladium catalysis, axial chirality, central chirality, simultaneous control, allene

Introduction

As we know, chirality is a very common phenomenon existing in nature. For example, the left hand is a non-superimposable mirror image of the right hand. At the molecular level, chirality is vital for understanding life on the earth and developing new chiral compounds for the treatment of disease. In addition to the extensively studied central chirality, molecules may also contain axial, planar, and helical chiralities, *etc*. (Figure 1A).[1] Of current interest, allenes are a class of very unique compounds with an axial chirality spreading over these intrinsic three carbon atoms. Among different chiralities listed in Figure 1, allenes have also become of essential interest for human beings due to their unique physical,[2] biological,[3] and chemical properties:[4] So far, about 220 natural products comprising an allenic structure are known,[3,5] some of such typical allenes are listed in Figure 1B; in addition, allene units have been observed to be important in (potential) drug molecules;[6] scientists have also demonstrated the beauty of allene-based transformations, especially, the axial chirality transfer-based reactions have provided an essential alternative solution in asymmetric synthesis.[7] As shown in Figure 1, many of these potentially useful compounds contain axial or central chiralities or both.[6-11] Thus, the development of new strategies for efficient highly stereoselective preparation of such allenes with both axial and central chiralities is of high academic interest and fundamental importance for science of chirality and synthesis as well as development of molecules with medicinal interest.

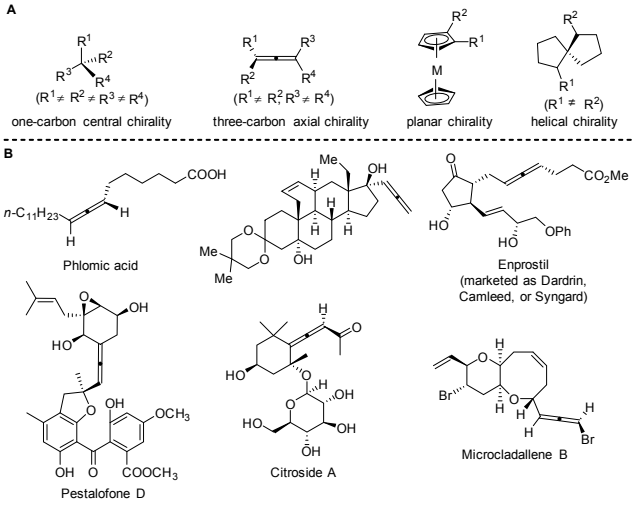


Figure 1 (A) Different chiralities of molecules; (B) Natural allenes, and (potential) drug molecules bearing allenes.

Catalytic enantioselective syntheses are always the most atom-economic and efficient method in asymmetric synthesis. Recently, transition metal-catalyzed enantioselective constructions of the axial chirality of allenes from 2-bromo-1,3-butadienes (path a),[12] racemic propargylic carbonates (path b),[13] stereo-defined enol triflates (path c),[14] conjugated enynes (path d),[15] and ketenes (path e),[16] *etc*.,[17] have been developed; in addition, Maruoka and List *et al.* applied the nucleophilic 1,2-addition of allenyl anions with aromatic imines or aldehydes (path f);[18,19] Feng *et al.* developed the chiral Lewis acid-catalyzed rearrangement of propargylic cyclic enol ethers (path g) or conjugate addition of malonate with stereo-defined 1-aryl-1-en-3-ynes (path h);[20,21] These strategies show a great potential with very promising diastereoselectivities in some cases (Scheme 1A). On the other hand, racemic 2,3-allenol derivatives have successfully been applied for the generation of optically active allenes with a central or axial chirality under the catalysis of a palladium catalyst via the "allene-asymmerization" strategy with different chiral ligands (Scheme 1B, eqs. 1 and 2).[22-26] However, no report has been disclosed with the simultaneous control of axial chirality and central chirality with such a strategy since different working principles are usually required for different chiralities.[27] Within such a protocol, we would have to clarify (1) the echoing between the central chirality and axial chirality on the enantioselectivity for both these two types of chirality, as well as (2) match and mis-match cases. Here, we wish to present the first example of catalytic asymmetric allenylation of an array of different useful nucleophiles such as bis(phenylsulfonyl)methane, fluorobis(phenylsulfonyl)methane, and malonate with one pair of diastereomer of the racemic secondary 1,3-disubstituted 2,3-allenylic acetates, providing an efficient access to optically active allenes bearing both central and axial chiralities with a single chiral ligand (Scheme 1C, eq. 3).

Results and Discussion

After some trials and errors, we started to work on the catalytic asymmetric reaction of racemic diastereoisomer (R_a*,*S**)-**1a**, which was prepared conveniently from the corresponding termi-

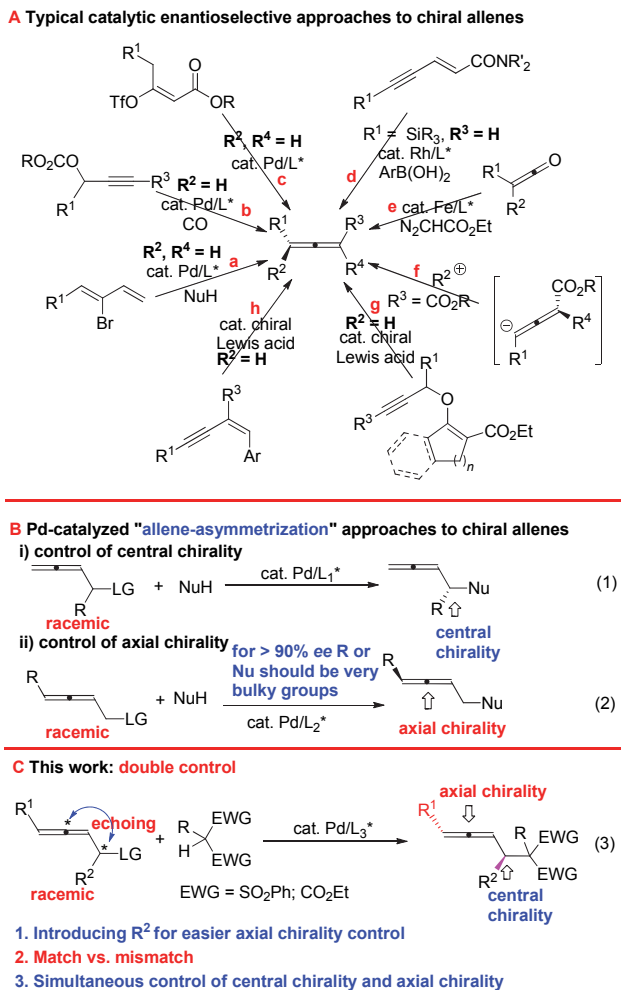
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† Dedicated to Professor Xiyan Lu on the occasion of his 90th birthday.

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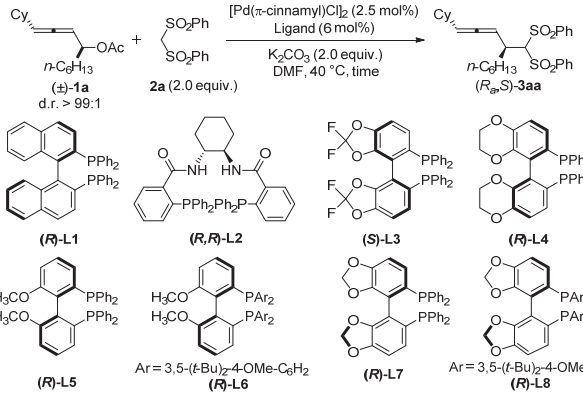
Scheme 1 Catalytic enantioselective syntheses of allenes



nal propargylic alcohol and aldehyde with different amines[28] followed by separation of the required diastereomer and esterification, with bis(phenylsulfonyl)methane (**2a**) in DMF with K_2CO_3 as the base. After a careful systematic screening of commercially available chiral bidentate bisphosphine ligands (Table 1, entries 1—8), (*R*)-DTBM-SEGPHOS (**L8**) has been identified to be the best in terms of both yield and stereoselectivity for the reaction in DMF. Furthermore, the yield has been improved with an even better *ee* by using anhydrous K_2CO_3 (heated at 500 °C for 8 h in a Muffle furnace) as the base and NMP as the solvent at 30 °C (Table 1, entry 9).

With the optimized reaction conditions, we explored the scope of substrates and found that an array of different 2,3-allenylic acetates could be efficiently transformed into the corresponding products with an excellent d.r. and enantioselectivity in moderate to high yields (Table 2). A very diverse range of substituents (R^1) including cyclic alkyl (**1a**, **1b**, **1h**, **1j**—**1l**), primary alkyl (**1c**, **1i**, **1m**), and electron-donating (**1e**) or electron-withdrawing aryl (**1f**, **1g**) have well been tolerated for this asymmetric allenylation reaction; synthetically attractive functional groups including a C=C bond (**1b**), OTBS (**1j**), and OAc (**1k**) on the R^2 substituents are also compatible, providing the desired products with excellent *ee*. In addition, the nucleophiles such as bis(phenylsulfonyl)methane (**2a**), fluorobis(phenylsulfonyl)methane (**2b**), and diethyl malonate (**2c**) all made no difference to the reactivity and selectivity. Gram scale synthesis was also realized easily affording desired product (R_a,*S*)-**3hc** with 98% *ee* (Table 2, entry 10). The absolute configuration of the product was determined by X-ray single crystal diffraction study of (R_a,*S*)-**3da** (Figure 2).[29]

Table 1 Optimization of the reaction conditions[a]



Entry	Ligand	Time/h	Recovery/%	(R_a,*S*)-**3aa**		
				NMR yield[b]/%	d.r.[b]	*ee*[c]/%
1	(*R*)-**L1**	14	35	51	>99:1	88.6
2	(*R*,*R*)-**L2**	48	88	0	—	—
3	(*S*)-**L3**	12	0	68	98:2	–91.0
4	(*R*)-**L4**	24	24	59	>99:1	92.9
5	(*R*)-**L5**	14	34	53	>99:1	94.3
6	(*R*)-**L6**	37	17	30	>99:1	64.0
7	(*R*)-**L7**	13	18	64	>99:1	94.1
8	(*R*)-**L8**	12	0	67	>99:1	94.7
9[d]	**(*R*)-L8**	**8.5**	**0**	**76**	**>99:1**	**96.9**

[a] Reagents and conditions: (±)-**1a** (0.2 mmol), **2a** (0.4 mmol), [Pd(π-cinnamyl)Cl]$_2$ (0.005 mmol), ligand (0.012 mmol), K_2CO_3 (0.4 mmol), and DMF (2 mL) at 40 °C. [b] Determined by ^{1}H NMR analysis using mesitylene as the internal standard. [c] Determined by chiral HPLC analysis after column chromatographic separation on silica gel. [d] The reaction was carried out with anhydrous K_2CO_3 in NMP at 30 °C.

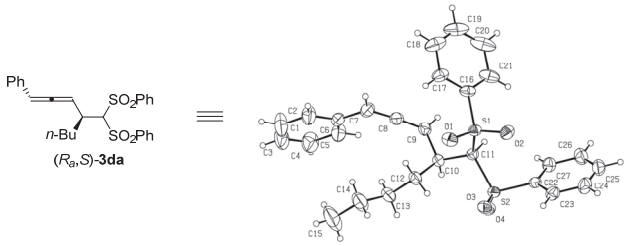


Figure 2 ORTEP representation of (R_a,*S*)-**3da**.

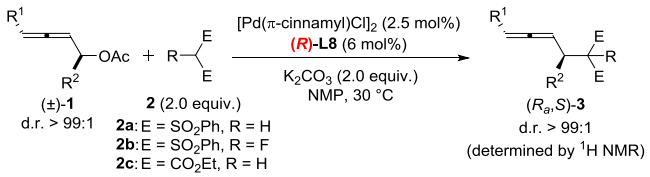
When the opposite enantiomer of chiral ligand (*R*)-**L8**, *i.e.*, (*S*)-**L8**, was applied under the optimized conditions, both the central chirality and axial chirality have been inverted to afford the enantiomeric products (S_a,*R*)-**3** with similar yields and *ee*s (Table 3).

In order to unveil any synergetic effect between two different types of chiralities, we then deliberately prepared 2,3-allenylic acetates bearing merely an axial chirality (**1q**) or a central chirality (**1r**) and explored their reactions with diethyl malonate (**2c**) under the standard conditions (Scheme 2A, eqs. 4 and 5). Interestingly, the enantioselectivity is merely 60% or 89%, respectively, confirming that the construction of a central chirality is much easier than that of axial chirality[22-26] and there is indeed a remarkable synergistic effect between them. In order to unveil the reaction mechanism, all four possible stereoisomers of 2,3-allenyl acetate **1h** have been prepared by the acetylation of chiral 2,3-allenol prepared according to the known procedure[28] and treated with nucleophile **2c** under the optimized conditions (Scheme 2B). (R_a,*S*)-**1h** and (S_a,*R*)-**1h** both gave the product (R_a,*S*)-**3hc** with high

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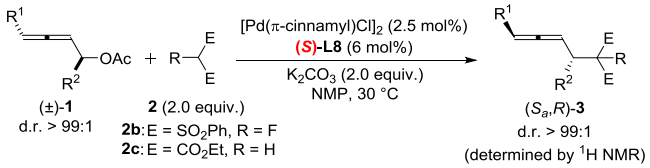
Table 2 The reaction of different racemic 2,3-allenyl acetate (R_a*,S*)-**1** with nucleophiles[a]



Entry	1		2	t/h	3	
	R^1	R^2			Yield[b]/%	ee[c]/%
1	Cy	*n*-C_6H_{13} (**1a**)	**2a**	12	77 (R_a,S)-**3aa**	96
2	Cy	 (**1b**)	**2a**	12	70 (R_a,S)-**3ba**	98
3	*n*-Bu	*n*-C_6H_{13} (**1c**)	**2a**	17	70 (R_a,S)-**3ca**	97
4	Ph	*n*-Bu (**1d**)	**2a**	15	55 (R_a,S)-**3da**	98
5	*p*-MeC_6H_4	*n*-C_6H_{13} (**1e**)	**2a**	12	65 (R_a,S)-**3ea**	97
6	*p*-BrC_6H_4	*n*-C_6H_{13} (**1f**)	**2a**	18	60 (R_a,S)-**3fa**	97
7	*p*-ClC_6H_4	*n*-C_7H_{15} (**1g**)	**2a**	18	55 (R_a,S)-**3ga**	98
8	Cy	*n*-C_7H_{15} (**1h**)	**2b**	12	81 (R_a,S)-**3hb**	97
9	Cy	*n*-C_7H_{15} (**1h**)	**2c**	12	74 (R_a,S)-**3hc**	98
10[d]	Cy	*n*-C_7H_{15} (**1h**)	**2c**	12	69 (R_a,S)-**3hc**	98
11	*n*-C_7H_{15}	*n*-C_6H_{13} (**1i**)	**2c**	16	80 (R_a,S)-**3ic**	98
12	Cy	 (**1b**)	**2c**	15	70 (R_a,S)-**3bc**	98
13	Cy	 (**1j**)	**2c**	14	75 (R_a,S)-**3jc**	97
14	Cy	(**1k**)	**2c**	20	77 (R_a,S)-**3kc**	98
15		*n*-C_9H_{19} (**1l**)	**2c**	17	78 (R_a,S)-**3lc**	>99
16	Ph 	*n*-C_9H_{19} (**1m**)	**2c**	16	83 (R_a,S)-**3mc**	96

[a] Reagents and conditions: (±)-**1** (1.0 mmol), **2** (2.0 mmol), [Pd(π-cinnamyl)Cl]$_2$ (0.025 mmol), (*R*)-**L8** (0.06 mmol), K_2CO_3 (2.0 mmol), and NMP (10 mL) at 30 °C. [b] Isolated yield. [c] Determined by chiral HPLC analysis. [d] Reaction was conducted on 1 g scale.

Table 3 The reaction of different racemic 2,3-allenyl acetate (R_a*,S*)-**1** with nucleophiles using (*S*)-**L8** as the ligand[a]



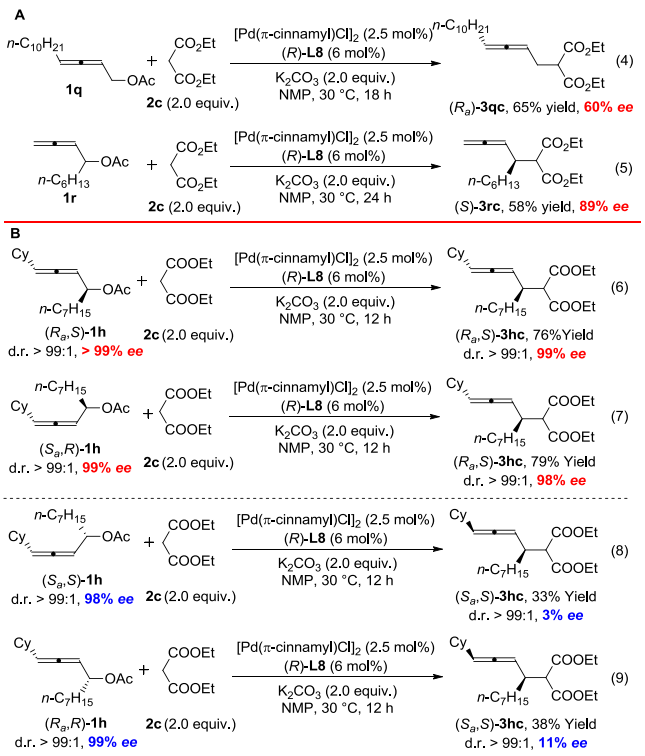
Entry	1		2	t/h	3	
	R^1	R^2			Yield[b]/%	ee[c]/%
1	Cy	*n*-C_7H_{15} (**1h**)	**2b**	12	78 (S_a,R)-**3hb**	97
2	Cy	*n*-C_7H_{15} (**1h**)	**2c**	12	73 (S_a,R)-**3hc**	99
3	*n*-C_7H_{15}	*n*-C_6H_{13} (**1i**)	**2c**	16	78 (S_a,R)-**3ic**	97
4	Cy	 (**1j**)	**2c**	14	75 (S_a,R)-**3jc**	97
5	Cy	 (**1k**)	**2c**	20	75 (S_a,R)-**3kc**	98
6	Ph 	*n*-C_9H_{19} (**1m**)	**2c**	16	81 (S_a,R)-**3mc**	98

[a] Reagents and conditions: (±)-**1** (1.0 mmol), **2** (2.0 mmol), [Pd(π-cinnamyl)Cl]$_2$ (0.025 mmol), (*S*)-**L8** (0.06 mmol), K_2CO_3 (2.0 mmol), and NMP (10 mL) at 30 °C. [b] Isolated yield. [c] Determined by chiral HPLC analysis.

yield, excellent diastereo- and enantioselectivity with (*R*)-**L8** as the ligand, indicating that complete racemization occurred for (S_a,R)-**1h**. However, (S_a,S)-**1h** and (R_a,R)-**1h** both gave the product (S_a,S)-**3hc** with low yield and poor enantioselectivity, although the diastereoiosmeric purity is still excellent, along with other unidentified byproducts.

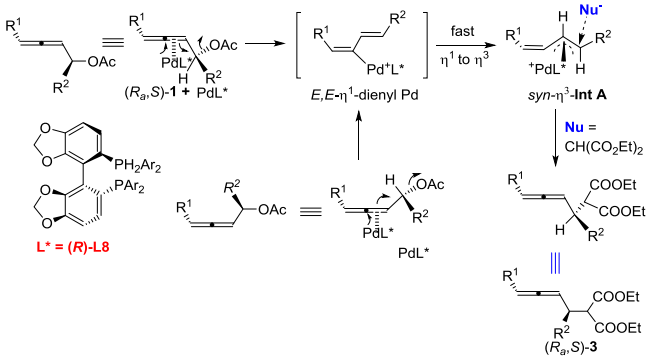
Based on these data in this studies and the X-ray diffraction

Scheme 2 Echoing between the central chirality and axial chirality (A) Cooperative effect; (B) Reactions of all four stereoisomers



data collected by Ogasawara/Takahashi[30] and Benyunes,[31] we proposed a rationale for the nature of the enantioselectivity for the generation of both the central chirality and axial chirality with (*R*)-**L8** as the ligand (Scheme 3): it is known that such S_N2'-type oxidative addition requires the attack of PdL* from the back side of the C—OAc bond,[32] thus, for a better understanding, all the four diastereomers of compound **1** are shown in a conformation with the C—OAc bond pointing backward for the front side attack of the PdL* species. Through such *anti*-oxidative addition, (R_a,S)-**1** or (S_a,R)-**1** would generate the same most favored *E*,*E*-η^1-dienyl Pd,[31] which would immediately yield the most favored delocalized intermediate *syn*-η^3-**Int A**[30] (for the Chem3D-based molecular model with both R^1 and R^2 being Me, see the separate SI file) via the σ-π rearrangement involving the coordination of the C=C bond connected to R^2 with Pd.[12,24,33] The front side of the carbon atom connected to the R^2 group in *syn*-η^3-**Int A** is blocked by (*R*)-**L8**, thus, attack of the carbon nucleophile from the back side of the π-allylic Pd unit on this specific carbon atom would form

Scheme 3 Rationale for the observed phenomena – simultaneous construction of the axial and central chirality: in the step of S_N2'-type oxidative addition, the PdL* is attacking from the back of the OAc group



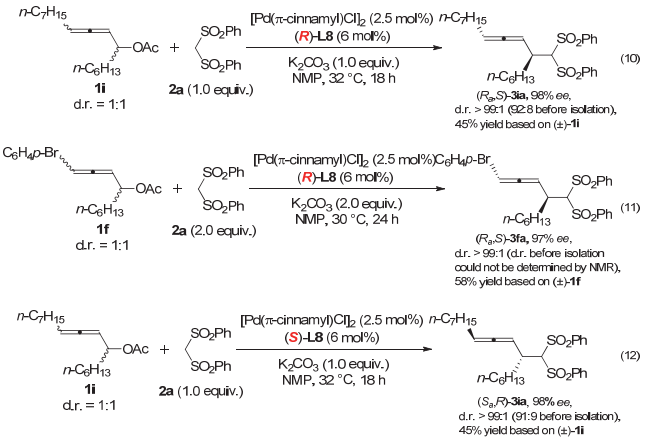
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the central chirality in *S* configuration as well as generate the allene unit with the C-C(Nu)-R^2 pointing downward generating the axial chirality in *R* configuration (see also the Chem3D-based molecular model with both R^1 and R^2 being Me in the separate SI. The geometry was fully optimized by B3LYP method with the Gaussian 09 software package. The 6-31G(d) basis set was used for C, H, O and P atoms and LANL2DZ basis set with effective core potential (ECP) was used for Pd atom.), leading to (R_a,*S*)-**3**.

In fact, based on these observations, kinetic resolution of palladium-catalyzed asymmetric allenylation using the 1 : 1 diastereoisomeric 2,3-allenyl acetates has been successfully developed by further optimization of the reaction parameters (Scheme 4, eqs. 10 and 11). The opposite enantiomers (S_a,*R*)-**3** could also be expediently prepared by switching the ligand from (*R*)-**L8** to (*S*)-**L8** under the optimized conditions with similar yield, diastereo- and enantioselectivity after isolation (Scheme 4, eq. 12).

Scheme 4 Kinetic resolution of asymmetric allenylation using the 1 : 1 diastereoisomeric 2,3-allenyl acetates **1** with (*R*)- or (*S*)-**L8**

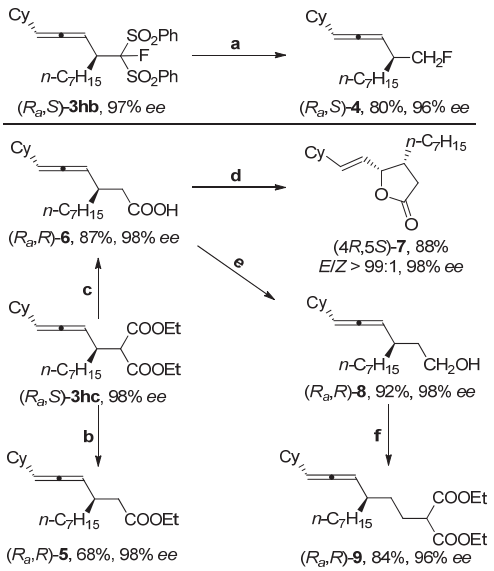


These allenes with both axial and central chirality are very useful building blocks for the synthesis of compounds containing these two types of stereocentres (Scheme 5). As we know, monofluorinated compounds are of great importance to bioisostere-based drug design,[34] thus (R_a,*S*)-**3hb** was treated with activated Mg turnings in MeOH/THF to remove the sulfonyl groups[35] to afford the monofluoromethylated allene product (R_a,*S*)-**4** in 80% yield and 96% *ee*. Direct removal of one of the ester groups in (R_a,*S*)-**3hc** was achieved by Krapcho decarboxylation,[36] providing 4,5-allenoate (R_a,*R*)-**5** in 68% yield with 98% *ee*; sequential alkaline hydrolysis followed by acidic decarboxylation of (R_a,*S*)-**3hc** afforded 4,5-allenoic acid (R_a,*R*)-**6** in 87% yield; gold-catalyzed cyclization of (R_a,*R*)-**6** has been carried out, affording γ-butyrolactone (4*R*,5*S*)-**7**, a key building block in the related lactone natural product synthesis,[37] with an excellent axial-to-central chirality transfer (98% *ee*); reduction of (R_a,*R*)-**6** with $LiAlH_4$ afforded 4,5-allenol (R_a,*R*)-**8** in 92% yield without any loss of enantiopurity (98% *ee*); the treatment of (R_a,*R*)-**8** with I_2 gave the corresponding 2,3-allenylic iodide, which underwent nucleophilic substitution with the diethyl malonate anion, providing (R_a,*R*)-**9** in 84% yield and 96% *ee*. In principle, such useful allenoic acids, allenoates, allenyl acetates, and allenols with both a remote axial and a central chirality could be easily achieved by repeatedly following these simple transformations from (R_a,*S*)-**3hc**.

Conclusions

In conclusion, our observation constitutes an echoing between the axial chirality and central chirality for a novel catalytic protocol for simultaneous construction of both axial chirality and central chirality in allenes using a single chiral ligand, providing a variety of structurally diversified optically active allenes containing both axial chirality and central chirality with functionalities of essential potentials such as acid, ester, malonate, hydroxyl, and fluoromethyl groups under mild conditions. We anticipate that this study will open the general interest for the development of catalytic technology for simultaneous asymmetric control of different types of chiralities and simplify the construction of optically active complex bioactive compounds. Further studies on this topic are being conducted in this laboratory.

Scheme 5 Synthetic applications



Reagents and conditions: (a) Mg (20 eq.), MeOH/THF, 0 °C—rt, 3 h; (b) LiCl (3.0 eq.), H_2O (4.0 eq.), DMSO, 180 °C, 4 h; (c) (1) NaOH (4 eq.), MeOH/H_2O, 100 °C, 2.5 h; (2) AcOH, 120 °C, 41 h; (d) Au(PPh_3)Cl (5 mol%), AgOTs (5 mol%), $CHCl_3$, –20 °C, 12 h; (e) $LiAlH_4$ (2.0 eq.), Et_2O, 0 °C, 3 h; (f) (1) PPh_3 (1.2 eq.), imidazole (1.2 eq.), I_2 (1.2 eq.), DCM, rt, 1 h; (2) NaH (1.2 eq.), $CH_2(CO_2Et)_2$ (2.0 eq.), DMF, rt, 12 h.

Experimental

General Information. ^{1}H NMR, ^{13}C NMR, and ^{19}F NMR spectra were recorded in $CDCl_3$ using a 300 MHz NMR spectrometer (^{1}H at 300 MHz, ^{13}C at 75 MHz, ^{19}F at 282 MHz). Mass spectra (MS) and high-resolution mass spectra (HRMS) were carried out using electron impact ionization (EI) or electron spray ionization (ESI). Anhydrous K_2CO_3 was heated at 500 °C for 6 h in a Muffle furnace and stored in a glove box. (*R*)- and (*S*)-**L8** were purchased from Strem Chemicals. Toluene, dioxane, tetrahydrofuran (THF), and diethyl ether (Et_2O) were freshly distilled from Na wire/benzophenone before use. 1-Methyl-2-pyrrolidinone (NMP) was distilled under vacuum from CaH_2 before use. Chloroform ($CHCl_3$) and dichloromethane (DCM) were freshly distilled from CaH_2 before use. Diethyl malonate was purchased from TCI Chemicals. Other commercially available reagents were purchased and used without further purification.

Typical procedure for the synthesis of (R_a,*S*)-3aa (Table 2, entry 1). To a dry Schlenk tube were added [Pd(π-cinnamyl)Cl]$_2$ (0.0134 g, 0.025 mmol), (*R*)-**L8** (0.0709 g, 0.06 mmol), and K_2CO_3 (0.2763 g, 2 mmol) in the glove box. Then NMP (5 mL) was added under nitrogen atmosphere. After the reaction was stirred in an oil bath preheated at 30 °C for 30 min, (±)-**1a** (0.2781 g, 1 mmol)/NMP (5 mL) and bis(phenylsulfonyl)methane (**2a**) (0.6105 g, 2 mmol) were added to the reaction mixture with stirring. After being stirred for 12 h at 30 °C, the reaction was complete as monitored by TLC, ethyl acetate (30 mL) was added. The resulting

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mixture was washed with brine (20 mL×3). The aqueous layer was extracted with Et_2O (20 mL). The combined organic layer was dried over anhydrous Na_2SO_4. After filtration and evaporation, the residue was purified by flash column chromatography (eluent: petroleum ether (30—60 °C)/ethyl acetate, *V*/*V*=8/1) on silica gel to afford (R_a,*S*)-**3aa** (0.3965 g, 77%) as an oil: 96% *ee* (HPLC conditions: Chiralcel OZ-H column, *V*(*n*-hexane)/*V*(*i*-PrOH)=90/10, 0.5 mL/min, λ=214 nm, t_R(major)=23.7 min, t_R(minor)=50.9 min); $[\alpha]_D^{20}$ –87.5 (*c* 1.03, $CHCl_3$); ^{1}H NMR (300 MHz, $CDCl_3$) δ: 7.92 (d, *J*=7.2 Hz, 2H, ArH), 7.85 (d, *J*=7.2 Hz, 2H, ArH), 7.64 (t, *J*=7.5 Hz, 2H, ArH), 7.57—7.42 (m, 4H, ArH), 5.51—5.40 (m, 1H, =CH), 5.12 (td, J_1=6.2 Hz, J_2=1.7 Hz, 1H, =CH), 4.67 (d, *J*=1.2 Hz, 1H, CH), 3.09—2.96 (m, 1H, CH), 2.01—1.73 (m, 3H, CH_2 and CH), 1.72—1.52 (m, 5H, 5H from Cy), 1.46—0.90 (m, 13H, 5H from Cy and CH_2×4), 0.86 (t, *J*=6.9 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ: 202.7, 140.1, 138.4, 134.2, 134.0, 129.1, 128.9, 128.74, 128.70, 98.7, 92.7, 87.4, 40.4, 36.8, 32.8, 31.4, 29.8, 28.4, 28.1, 25.72, 25.66, 22.3, 13.8; IR (neat) *v*: 3066, 2924, 2852, 1960, 1584, 1478, 1463, 1448, 1334, 1312, 1291, 1158, 1079 cm^{-1}; MS (70 eV, EI) *m/z* (%): 514 (M^+, 1.46), 77 (100); HRMS calcd for $C_{29}H_{38}O_4S_2$ [M^+]: 514.2212, found: 514.2216.

Supporting Information

The supporting information for this article is available on the WWW under https://doi.org/10.1002/cjoc.201800090.

Acknowledgement

Financial support from the National Natural Science Foundation of China (21690063) and National Basic Research Program (2015CB856600) of China is greatly appreciated. We thank Mr. Yang Tang in our group for reproducing the result presented in entry 2 in Table 3. Shengming Ma is a Qiu Shi Adjunct Professor at Zhejiang University.

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Manuscript received: February 23, 2018
Manuscript accepted: February 24, 2018
Accepted manuscript online: February 28, 2018
Version of record online: March 15, 2018

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